- $\forall\ k \geq 1$,

$$\text{Corr}(y_t, y_{t-k}) = \frac{\text{Cov}(y_t, y_{t-k})}{\sqrt{\text{Var}(y_t)\,\text{Var}(y_{t-k})}}$$

$$\text{Corr}(y_t, y_{t-k}) = \frac{\gamma_x(k)}{\text{Var}(y_t)}$$

by stationarity of $\{y_t\}$, and $\text{Var}(y_t) = \sigma_x^2 + \sigma_e^2$
by dividing by σ_x^2 we have:

$$\text{Corr}(y_t, y_{t-k}) = \frac{\gamma_x(k)/\sigma_x^2}{1 + \frac{\sigma_e^2}{\sigma_x^2}}$$

finally:

$$\text{Corr}(y_t, y_{t-k}) = \frac{\rho_k}{1 + \frac{\sigma_e^2}{\sigma_x^2}}$$